MA10207 Analysis 1 - Semester 1, 2021/22

Problem Sheet Week 8 - Solutions

1. Investigate the convergence or divergence of the series $\sum_{n=1}^{\infty} a_n$, when

$$a. \quad a_n = \sqrt{n+1} - \sqrt{n}, \quad b. \quad a_n = \frac{\sqrt{n+1} - \sqrt{n}}{n}$$

Answer. a. It holds

$$a_n = \sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \geq \frac{1}{3\sqrt{n}}$$

for any $n \geq 1$. Since $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges.

b. It holds

$$a_n = \frac{1}{n} \frac{1}{\sqrt{n+1} + \sqrt{n}} \leq \frac{1}{n\sqrt{n}}, \quad \forall n \geq 1.$$

Since the series $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}}$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

2. Use the comparison test to decide whether the following series converge or diverge:

$$a. \quad \sum_{n=1}^{\infty} \frac{1}{1+n^5}, \quad b. \quad \sum_{n=1}^{\infty} \frac{n^4+1}{n^5+3}, \quad c. \quad \sum_{n=1}^{\infty} \frac{2}{\binom{3n+2}{3n}}.$$

Answer. a. Since

$$0 \leq \frac{1}{1+n^5} \leq \frac{1}{n^5}$$

and since $\sum_{n=1}^{\infty} \frac{1}{n^5}$ converges, by comparison it follows $\sum_{n=1}^{\infty} \frac{1}{1+n^5}$ converges.

b. We write

$$\frac{n^4+1}{n^5+3} = \frac{1}{n} \frac{1 + \frac{1}{n^4}}{1 + \frac{3}{n^5}} \geq \frac{1}{n} \frac{1}{1 + \frac{3}{n^5}} \geq \frac{1}{4n},$$

where we used the facts that $1 + \frac{1}{n^4} \geq 1$, and $1 + \frac{3}{n^2} \leq 4$. Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, by comparison it follows $\sum_{n=1}^{\infty} \frac{n^4+1}{n^5+3}$ diverges.

c. We have

$$o \leq \frac{2}{\binom{3n+2}{3n}} = \frac{4(3n)!}{(3n+2)!} = \frac{4(3n)!}{(3n+2)(3n+1)(3n)!} = \frac{4}{(3n+2)(3n+1)}.$$

Hence

$$0 \leq \frac{2}{\binom{3n+2}{3n}} \leq \frac{4}{9n^2}, \quad \forall n \geq 1.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent, by comparison also the series $\sum_{n=1}^{\infty} \frac{2}{\binom{3n+2}{3n}}$ is convergent.

3. Use the Ratio Test to investigate the convergence of the following series:

$$a.\quad \sum_{n=1}^{\infty} \frac{2^n}{n^6}, \quad b.\quad \sum_{n=1}^{\infty} \frac{n^3}{n!}, \quad c.\quad \sum_{n=1}^{\infty} \frac{3^n}{2^n+1}.$$

Answer. Since

$$\lim_{n\to\infty} \frac{2^{n+1}}{(n+1)^6} \frac{n^6}{2^n} = \lim_{n\to\infty} 2(1+\frac{1}{n})^6 = 2 > 1,$$

by the Ratio Test we conclude that $\sum_{n=1}^{\infty} \frac{2^n}{n^6}$ diverges.

b. Since

$$\lim_{n\to\infty} \frac{(n+1)^3}{(n+1)!} \frac{n!}{n^3} = \lim_{n\to\infty} \frac{1}{n+1}(1+\frac{1}{n})^3 = 0 < 1,$$

by the Ratio Test we conclude that $\sum_{n=1}^{\infty} \frac{n^3}{n!}$ converges.

c. Since

$$\lim_{n\to\infty} \frac{3^{n+1}}{2^{n+1}+1} \frac{2^n+1}{3^n} = \lim_{n\to\infty} \frac{3}{2} \frac{(1+\frac{1}{2^n})}{(1+\frac{1}{2^{n+1}})} = \frac{3}{2} > 1,$$

by the Ratio Test we conclude that $\frac{3^n}{2^n+1}$ diverges.

4. Let $(a_n)_n$ be a sequence of positive numbers such that

$$\lim_{n\to\infty} \frac{a_n}{n} = 2.$$

Prove that $\sum_{n=1}^{\infty} \frac{2a_n}{1+a_n^3}$ is convergent.
Prove that $\sum_{n=1}^{\infty} \frac{2a_n}{1+a_n^2}$ is divergent.

Answer. Since $\lim_{n\to\infty} \frac{a_n}{n} = 2$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$

$$|\frac{a_n}{n} - 2| < 1 \quad \Leftrightarrow \quad n < a_n < 3n.$$

Hence

$$0 \leq \frac{2a_n}{1+a_n^3} \leq \frac{6n}{1+n^3} \leq \frac{6}{n^2}, \quad \forall n \geq N.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent, by the comparison principle we conclude that $\sum_{n=1}^{\infty} \frac{2a_n}{1+a_n^3}$ is convergent.

Let us now consider $\sum_{n=1}^{\infty} \frac{2a_n}{1+a_n^2}$. In this case we have

$$\frac{2a_n}{1+a_n^2} \geq \frac{2n}{1+9n^2} \geq \frac{n}{9n^2} = \frac{1}{9n}.$$

Since the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, the comparison principle yields that also $\sum_{n=1}^{\infty} \frac{2a_n}{1+a_n^2}$ is divergent.

Homework.

1. Determine whether the following series are convergent or divergent

$$a. \quad \sum_{n=1}^{\infty} \frac{n^n}{(2n)!} \qquad b. \quad \sum_{n=1}^{\infty} \frac{1+\cos n}{n^4}$$

[You may assume $-1 \le \cos x \le 1$ for all real x.]

Answer. a. Let $a_n = \frac{n^n}{(2n)!}$. We have $a_n \ge 0$ for all $n \ge 1$, and

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{(n+1)^{n+1}}{(2(n+1))!}\frac{(2n)!}{n^n} = (\frac{n+1}{n})^n(n+1)\frac{(2n)!}{(2n+2)(2n+1)(2n)!} \\ &= (1+\frac{1}{n})^n \frac{n+1}{(2n+2)(2n+1)}.\end{aligned}$$

Recall that $\lim_{n\to\infty}(1+\frac{1}{n})^n = e$. Hence

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = 0.$$

By the Ratio Test we conclude that $\sum_{n=1}^{\infty} \frac{n^n}{(2n)!}$ is convergent.

b. We observe that

$$0 \le \frac{1+\cos n}{n^4} \le \frac{2}{n^4}.$$

Since $\sum_{n=1}^{\infty} \frac{2}{n^4}$ converges, then by comparison the series $\sum_{n=1}^{\infty} \frac{1+\cos nx}{n^4}$ converges.

2. Determine whether the following series are convergent or divergent

$$a. \quad \sum_{n=0}^{\infty} \left(\frac{1}{n+2}\right)^n \qquad b. \quad \sum_{n=1}^{\infty} \frac{1}{\binom{4n}{3n}}.$$

Answer. a. Let $a_n = \left(\frac{1}{n+2}\right)^n$. We have $a_n \ge 0$ for all $n \ge 1$, and

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \left(\frac{n+2}{n+3}\right)^n \frac{1}{n+3} \\ &= \left(1-\frac{1}{n+3}\right)^n \frac{1}{n+3} \\ &= \left(1-\frac{1}{n+3}\right)^{n+3}\left(1-\frac{1}{n+3}\right)^{-3}\frac{1}{n+3}.\end{aligned}$$

We claim that

$$\lim_{n\to\infty}\left(1-\frac{1}{n+3}\right)^{n+3} = e^{-1}.$$

Indeed,

$$
\begin{aligned}
\lim_{n\to\infty}\left(1-\frac{1}{n+3}\right)^{n+3} &= \lim_{k\to\infty}\left(1-\frac{1}{k}\right)^{k} \\
&= \lim_{k\to\infty}[\left(1-\frac{1}{k}\right)^{-1}]^{-k} \\
&= \lim_{k\to\infty}[\left(1+\frac{1}{k-1}\right)^{k}]^{-1} = e^{-1}.
\end{aligned}
$$

Since

$$\lim_{n\to\infty}\left(1-\frac{1}{n+3}\right)^{-3}\frac{1}{n+3} = 0,$$

we conclude that

$$\lim_{n\to\infty}\frac{a_{n+1}}{a_n} = 0.$$

By the Ratio Test, we conclude that the series is convergent.

b. Let $a_n = \frac{1}{\binom{4n}{3n}} = \frac{(3n)!n!}{(4n)!}$. We have $a_n \geq 0$ for all $n \geq 1$ and

$$
\begin{aligned}
\frac{a_{n+1}}{a_n} &= \frac{(3(n+1))!\,(n+1)!}{(4(n+1))!}\cdot\frac{(4n)!}{(3n)!\,n!} \\
&= \frac{(n+1)!}{n!}\cdot\frac{(4n)!}{(4(n+1))!}\cdot\frac{(3(n+1))!}{(3n)!} \\
&= (n+1)\cdot\frac{1}{(4n+4)(4n+3)(4n+2)(4n+3)}\cdot(3n+3)(3n+2)(3n+1) \\
&= \frac{(n+1)(3n+3)(3n+2)(3n+1)}{(4n+4)(4n+3)(4n+2)(4n+3)}.
\end{aligned}
$$

Since

$$\lim_{n\to\infty}\frac{a_{n+1}}{a_n} = \frac{27}{256} < 1,$$

by the Ratio Test we conclude that $\sum_{n=1}^{\infty}\frac{1}{\binom{4n}{3n}}$ is convergent.

3. If $a_n \geq 0$ for all $n \in \mathbb{N}$, $\sum_{n=1}^{\infty} a_n$ converges and $(b_n)_{n\in\mathbb{N}}$ is monotone decreasing and bounded, prove that $\sum_{n=1}^{\infty} a_n b_n$ converges.

Answer. Since $(b_n)_{n\in\mathbb{N}}$ is monotone and bounded, $\lim_{n\to\infty} b_n = b$. Since $(b_n)_{n\in\mathbb{N}}$ is monotone decreasing and bounded, there exists $M > 0$ so that

$$0 \leq b_n - b \leq M, \quad \forall \quad n \geq 1.$$

We write

$$a_n b_n = a_n b + a_n(b_n - b).$$

The series $\sum_{n=1}^{\infty} a_n b = b \sum_{n=1}^{\infty} a_n$ converges since $\sum_{n=1}^{\infty} a_n$ converges. Let us now consider the series $\sum_{n=1}^{\infty} a_n(b_n - b)$. We have

$$0 \leq a_n(b_n - b) \leq a_n M \quad \forall \quad n \geq 1.$$

By the comparison test, we conclude that the series $\sum_{n=1}^{\infty} a_n(b_n - b)$ is convergent. Hence the series

$$\sum_{n=1}^{\infty} a_n b_n = \sum_{n=1}^{\infty} a_n b + \sum_{n=1}^{\infty} a_n(b_n - b)$$

converges, being the sum of two convergent series.

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